

data-model

Phase 1 — Data Model

Feature: Authored AI/IT Knowledge Areas · **Date:** 2026-09-07

Entities, their invariants, and the state each can be in. Field names are the shape, not a serialisation; the wire form is in [contracts/](#).

Two invariants govern everything below, and both exist because their absence was the measured defect:

- **I1 — One decision, many consumers.** The publication decision is computed once. Any entity’s visibility derives from that single computation. Two implementations that agree today are the D3 defect waiting to recur.
- **I2 — Absence is a state, not a null.** Every “missing” case is a named, reportable state with a reason. A synthesised value standing in for an absent one is how 814 mined areas acquired names they never had.

Area

The published unit: a named AI/IT subject.

Field	Notes
id	ULID. The identity. Minted by extraction; authored content binds to it. Never displayed as a title.
title	Human-readable. Non-empty is the publication rule (R3).
summary	What the subject covers.
tags	

Field	Notes
	Derived. Never null — an empty array instead , because a null breaks the field the interface filters on.
evidence_count	Passages supporting this area. Reported, never used as a publication filter (R2).
lessons	Ordered. May be empty only if the area is not published.
assessment	Zero or one.
corpus_revision	The revision this area was built from (FR-023a).

Invariants

- title non-empty is the title test for publication.
- An area appears in the catalogue **iff** its detail, lessons and assessment are servable (FR-018). Listing and openability are one condition.
- Withheld areas keep their identifier and reason, and disclose a derived name **when one is derivable** (FR-018b). Withholding an area is not a reason to withhold what it is called.
- Thinly-evidenced areas still report evidence_count (FR-005b).

States — the closed vocabulary, every one reportable by every surface:

State	Meaning	Detail route	In areas []
published	all tests pass	200	yes
no_title	mined proposal, never authored	refusal	no
no_evidence	no supporting passages	refusal	no
no_publication_review	authored, unreviewed	refusal + derived name	no
stale_publication_review			no

State	Meaning	Detail route	In areas[]
	review predates content	refusal + derived name	
no_materials_record	editorial: nothing attached	refusal	no
build_inconsistent	review present, materials absent — a build fault, not editorial	503	no

build_inconsistent is deliberately distinct (FR-018c): the author stage writes review and materials in one run, so their disagreement is a broken build. It answers 503 while the catalogue still answers 200 — one bad record must not take the catalogue down.

Lesson

Field	Notes
id, ord	ord drives one server-side sort. One order, used by list and by prev/next.
title, body	
estimated_minutes	Per lesson; summed to an area total.
materials[]	Zero or more.
anchors[]	Zero or more video anchors.
required	Whether it counts toward assessment availability.

Invariants

- Order is total and stable. Prev/next are computed **from the same sorted slice the list uses** — a second ordering is a second thing to disagree.

- The required set **MUST** be able to be a **proper subset** of the lessons; a model where they coincide cannot express the gate correctly and hides the “any lesson opens it” defect (R6).

Material

Field	Notes
kind	Closed vocabulary: illustration, diagram, scheme, graph, video. An unknown kind is a 400 , never an empty list.
caption	What it shows.
alt	Text alternative conveying content , not a file name.
ref	Where the asset is.

Invariant: a material whose asset is missing renders as *unavailable*, never as a broken element.

VideoAnchor

Field	Notes
chapter_id	Which recording.
start_millis	Measured from the segment sidecar. Not derivable from the passage registry, which carries only an end time.
end_millis	Optional.
transcript_anchor	The passage id to scroll to and mark.

Invariants

- Every anchor in a published lesson resolves to an existing recording **and** an existing passage (FR-016); an unresolvable anchor blocks publication of that lesson rather than reaching a learner.

- end is emitted **only when** `> t`. A malformed span is ignored by the consumer, so emitting one produces a broken link that renders correctly.
- The `#p-` fragment is emitted **only for a Crockford-shaped ULID**.
- An unservable chapter yields `href: null + unresolved_reason + intact_anchor + a count`. **No link is invented.**

Assessment

Field	Notes
<code>questions[]</code>	The bank.
<code>pass_percent</code>	Declared per area; shown before the attempt (FR-008b).
<code>required_lessons[]</code>	The availability condition.

Invariants

- Availability is decided **where content is served**, not where it is displayed (FR-012a).
- While unavailable, the response carries **zero** prompts, options and answers (FR-010, SC-009).
- An area may have lessons and **no** assessment. That is an honest state; a fabricated question bank is worse than a missing one.

Question

Field	On the taking path	In the result
<code>id, prompt, options[]</code>	yes	yes
<code>correct_choices[]</code>	absent by construction	yes
<code>explanation</code>	absent by construction	yes

Invariant: the taking projection is built as a map with **no answer field at all** — not as the domain type with fields omitted. Omission depends on every future field remembering to omit itself; absence does not.

Answers are identified **by ID, never by index** (FR-012), so presentation order cannot change what an answer means.

Attempt

Field	Notes
session	Opaque, header-supplied, one channel only (R7).
answers	By question id and choice id.
score, passed	
outcomes[]	Per question: correct?, correct answer, explanation.

Invariants

- A submitted result is **stable** — re-reading returns the same result (FR-011).
- Progress is mutated under a single lock in one operation; a read-then-write cycle silently drops a concurrent completion (R7).

Catalogue

Field	Notes
areas[]	Exactly the set whose detail answers 200.
held_back[]	id + reason for every withheld area.
reason_table	The closed vocabulary, in the same response.
total_areas	The population, so a filtered list never hides the denominator.
corpus_revision	What this was built from (FR-023a).

Invariant (I1, asserted end to end): areas[] is exactly the set whose href answers 200. This is a property a gate can check, and does.

Nothing becomes unreachable by being held back: ?
include=held_back returns full objects for those rows.